

Class Equation Short Tricks

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1 Type of Groups

We will use short-cut trick on the following type of groups of order:

- p^2
- p^3
- pq such that p and q are distinct primes and $p < q$ and $p \nmid q - 1$
- p -local groups

And groups of the following special types:

- D_n
- S_n

2 Short Tricks

2.1 For p^2

Group of order p^2

The group of order p^2 is always abelian. So, class equation is:

$$p^2 = \sum_i^{p^2} 1$$

2.2 For p^3

Group of order p^3

The non-abelian group of order p^3 has a center always of order p . So, class equation is:

$$p^3 = \sum_1^p 1 + (p^2 - 1)p$$

2.3 For pq

Group of order pq

If p and q are distinct primes such that $p < q$ and $p \nmid q - 1$, then the group of order pq has class equation:

$$pq = 1 + \sum_1^{\frac{q-1}{p}} p + \sum_1^{p-1} q$$

2.4 For D_n

Dihedral Group D_n

The dihedral group D_n has order $2n$. The class equation for D_n depends on whether n is even or odd:

$$\begin{aligned} \text{If } n \text{ is even: } 2n &= 1 + 1 + \sum_{i=1}^{\frac{n-2}{2}} 2 + \frac{n}{2} + \frac{n}{2} \\ \text{If } n \text{ is odd: } 2n &= 1 + n + \sum_{i=1}^{\frac{n-1}{2}} 2 \end{aligned}$$

3 Number of Conjugacy Classes

3.1 For p^2

Conjugacy Classes for p^2

The group of order p^2 is abelian, so it has p^2 conjugacy classes.

3.2 For p^3

Conjugacy Classes for p^3

The non-abelian group of order p^3 has a center of order p , so it has $p^2 + p - 1$ conjugacy classes.

3.3 For pq

Conjugacy Classes for pq

If p and q are distinct primes such that $p < q$ and $p \nmid q - 1$, then the group of order pq has $\frac{q-1}{p} + p$ conjugacy classes.

3.4 For D_n

Conjugacy Classes for D_n

The dihedral group D_n has the following number of conjugacy classes:

$$\begin{aligned} \text{If } n \text{ is even: } & \frac{n+6}{2} \\ \text{If } n \text{ is odd: } & \frac{n+3}{2} \end{aligned}$$

3.5 For S_n

Conjugacy Classes for S_n

The number of conjugacy classes is: $P(n) :=$ no. of different partitions of n .